Notebook Link: https://colab.research.google.com/drive/1yHH-yVKgJV9_XocclgN21q0HxHP7QIpA?usp=sharing

In [3]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import os
import shutil
import cv2
```

In [4]:
```
from google.colab import drive, files
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

In [ ]:
```
!unzip '/content/drive/My Drive/APS360/archive (2).zip' -d '/content/'
```

In [6]:
```
allowed_classes = [
    'Basketball', 'CricketShot', 'TennisSwing', 'Bowling', 'BaseballPitch',
    'JavelinThrow', 'Archery', 'PoleVault', 'SoccerPenalty', 'Biking',
    'Skiing', 'Fencing'
]

directories = ['/content/test', '/content/train', '/content/val']

# Loop through folder for each split and delete unwanted class folders
for dir in directories:
  for class_folder in os.listdir(dir):
    full_path = os.path.join(dir, class_folder)

    if class_folder not in allowed_classes:
        shutil.rmtree(full_path)
```

In [7]:
```
# Define constants
INPUT_DIR = '/content/'
OUTPUT_DIR = '/content/ucf_dataset_frames'
NUM_FRAMES_PER_VIDEO = 16
TARGET_IMG_SIZE = 224 # This variable is used for resizing again.

# Create output directories if they don't exist
```

```python
for split_dir in ['train', 'test', 'val']:
    os.makedirs(os.path.join(OUTPUT_DIR, split_dir), exist_ok=True)


def extract_frames(video_path):
  frames = []

  # Open video file
  cap = cv2.VideoCapture(video_path)

  # Return early if error occurs while tyring to open video file
  if not cap.isOpened():
    print(f"Error: could not open video {video_path}")
    return frames

  # See how much frames video currently has
  total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))


  # Uniformly sample NUM_FRAMES_PER_VIDEO frames
  interval = total_frames / NUM_FRAMES_PER_VIDEO
  sampling_indices = [int(i * interval) for i in range(NUM_FRAMES_PER_VIDEO)]

  # Loop through sampling indices
  for i, frame_idx in enumerate(sampling_indices):
    cap.set(cv2.CAP_PROP_POS_FRAMES, frame_idx)
    ret, frame = cap.read()

    # Resize frame to target size
    resized_frame = cv2.resize(frame, (TARGET_IMG_SIZE, TARGET_IMG_SIZE))

    # Frames are appended after resizing, typically uint8 in original 0-255 range.
    frames.append(resized_frame)

  # Close video file
  cap.release()

  return frames
```

In [8]:

```python
directories = ['train', 'val', 'test']
```

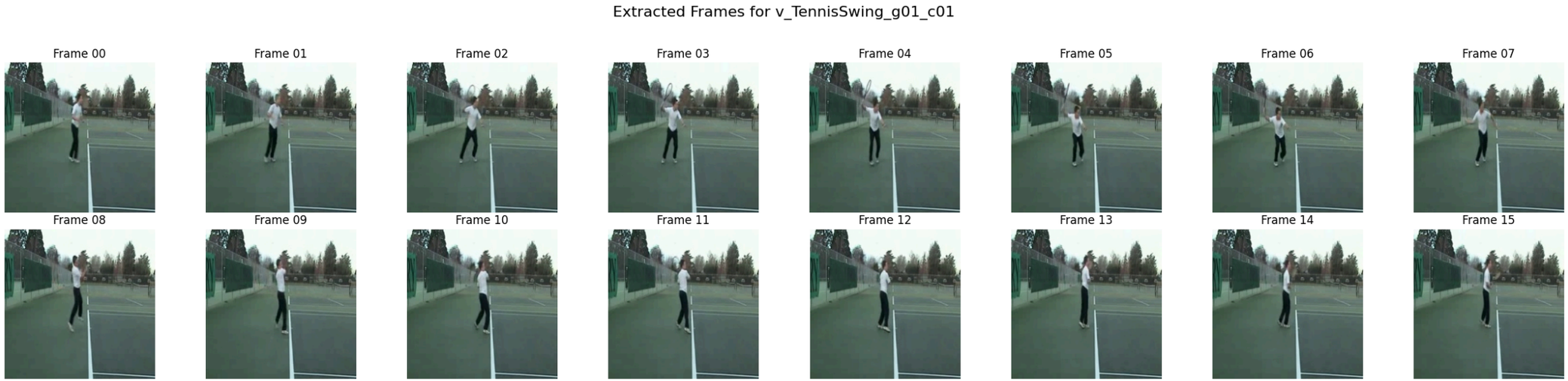
```
for split_dir in directories:
  # Get the input and output dir path for current split
  split_input_path = os.path.join(INPUT_DIR, split_dir)
  split_output_path = os.path.join(OUTPUT_DIR, split_dir)

  for class_name in allowed_classes:
    # Create the subdirectory for the current class in the output split directory
    input_class_path = os.path.join(split_input_path, class_name)
    output_class_path = os.path.join(split_output_path, class_name)
    os.makedirs(output_class_path, exist_ok=True)

    # Get all videos in the input split folder of current class
    video_files = [f for f in os.listdir(input_class_path)]

    for video_filename in video_files:
      # For every video get the sampled 16 frames
      video_path = os.path.join(input_class_path, video_filename)
      video_name_without_ext = os.path.splitext(video_filename)[0]
      extracted_frames = extract_frames(video_path)

      if len(extracted_frames) == 0:
        print(f"Skipping corrupted video: {video_path}")
        continue

      # Create a subfolder for this clip
      clip_output_path = os.path.join(output_class_path, video_name_without_ext)
      os.makedirs(clip_output_path, exist_ok=True)

      # For every frame add the frame # to file name and write to class subdirectory in output split directory
      for i, frame in enumerate(extracted_frames):
        # Save frames directly as they come from extract_frames (original resolution, uint8, 0-255 range).
        frame_filename = f"frame_{i:02d}.jpg"
        frame_output_path = os.path.join(clip_output_path, frame_filename)
        cv2.imwrite(frame_output_path, frame)
```

In [9]:

```
# Example video details from previous execution
example_split_dir = 'train'
example_class_name = 'TennisSwing'
example_video_name_without_ext = 'v_TennisSwing_g01_c01'

# Construct the base path to the frames
example_frames_dir = os.path.join(OUTPUT_DIR, example_split_dir, example_class_name, example_video_name_without_ext)
```

```
print(example_frames_dir)

plt.figure(figsize=(24, 6)) # Adjusted figure size for 8 frames per row

loaded_frames_count = 0
for i in range(NUM_FRAMES_PER_VIDEO):
  frame_filename = os.path.join(example_frames_dir, f"frame_{i:02d}.jpg")
  if os.path.exists(frame_filename):
    # Read and convert image from OpenCV format to matplotlib format
    frame = cv2.imread(frame_filename)
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    # Plot image
    plt.subplot(2, 8, i + 1)
    plt.imshow(frame_rgb)
    plt.title(f"Frame {i:02d}")
    plt.axis('off')
    loaded_frames_count += 1

  else:
      print(f"Warning: Frame file not found: {frame_filename}")


plt.suptitle(f"Extracted Frames for {example_video_name_without_ext}", fontsize=16)
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()
```

```
/content/ucf_dataset_frames/train/TennisSwing/v_TennisSwing_g01_c01
```



Example of a cleaned training sample

In [10]:
```
class_split_counts = {class_name: {'train': 0, 'val': 0, 'test': 0, 'total': 0} for class_name in allowed_classes}

for split_dir in ['train', 'val', 'test']:
  split_path = os.path.join(OUTPUT_DIR, split_dir)
  for class_name in allowed_classes:
    class_path = os.path.join(split_path, class_name)
    count = len([f for f in os.listdir(class_path) if os.path.isdir(os.path.join(class_path, f))])
    class_split_counts[class_name][split_dir] = count
    class_split_counts[class_name]['total'] += count

df_class_counts = pd.DataFrame.from_dict(class_split_counts, orient='index')
df_class_counts
```

Out[10]:

	train	val	test	total
Basketball	198	33	34	265
CricketShot	125	21	21	167
TennisSwing	124	21	21	166
Bowling	116	19	20	155
BaseballPitch	112	19	19	150
JavelinThrow	87	15	15	117
Archery	108	18	19	145
PoleVault	111	19	19	149
SoccerPenalty	102	17	18	137
Biking	100	17	17	134
Skiing	101	17	17	135
Fencing	83	14	14	111

In [11]:
```
df_class_counts.to_excel('class_counts.xlsx', index=True)
```

In [ ]:
```
!zip -r ucf_dataset_frames.zip /content/ucf_dataset_frames
```

In [13]:
```
files.download('ucf_dataset_frames.zip')
```

In [14]:
```
%%shell
jupyter nbconvert --to html /content/aps360_data_processing.ipynb
```

```
[NbConvertApp] Converting notebook /content/aps360_data_processing.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 1 image(s).
[NbConvertApp] Writing 2221491 bytes to /content/aps360_data_processing.html
```

Out[14]: